

Two ways to transcribe speech on a phone

Seven recordings, two transcription engines, fourteen runs, all fed at normal speaking speed and scored the same way. A second team measured the same Android engine independently against a published scoring protocol, and their figures appear alongside. We ran all of this after aligning our scoring with that protocol. Earlier figures came from a different alignment method, so they are not here.

On short recordings the two engines are level. Across five English dialogues of one to five minutes, Whisper averaged 2.8% word errors and the phone's built-in recogniser 3.3%, and the built-in engine was the more accurate of the two on the longest of them.

The gap opens on long recordings. On a 21-minute English narration Whisper made 8.0% errors against 10.0%, and on 17 minutes of German, 8.4% against 13.5%.

Whisper is also two to three times faster, and costs more to run. It transcribes at 2.1 to 2.6 times real time, while the built-in service will not take audio faster than someone spoke it. The price is a 240 MB model that ships with the app and stays in memory. The built-in engine ships nothing.

The two engines are Android's built-in recogniser, a system service that ships with the phone and needs no model of its own, and Whisper Small, OpenAI's open speech model, running locally on the CPU. A third route, a general-purpose language model with audio input, has not been re-measured under the current scoring, so it has no number here. The recordings are a 21:30 English narration, a 17:14 German conversation set, and five shorter English dialogues. Text-to-speech spoke all of them, which makes the script behind each one an exact reference to score against. Both engines received the audio at normal speaking speed, the only rate that matches how someone actually records. Coverage, meaning how much transcript came back against the reference, ran between 96% and 101% on every run below, so none of these numbers hides a shortened transcript.

ACCURACY AT NORMAL SPEED

RECORDING	LENGTH	WHISPER WER	ANDROID WER	WHISPER CER	ANDROID CER	WHICH IS AHEAD
English narration	21:30	8.0%	10.0%	2.7%	3.3%	Whisper, by 2.0 points
German conversations	17:14	8.4%	13.5%	4.4%	8.2%	Whisper, by 5.1 points
English dialogue	4:49	2.4%	1.7%	0.9%	0.8%	Android, by 0.7 points
English dialogue	3:12	2.4%	3.1%	0.8%	1.1%	Whisper, by 0.7 points
English dialogue	2:53	3.6%	4.3%	1.2%	2.1%	Whisper, by 0.7 points
English dialogue	2:04	2.4%	3.7%	0.6%	0.9%	Whisper, by 1.3 points
English dialogue	1:44	3.2%	3.9%	0.9%	0.9%	Whisper, by 0.7 points

Lower is better throughout. The second team's kit, which normalises text slightly differently, scores the two long recordings at 6.01% and 8.48% for Whisper and 8.79% and 11.75% for Android, against their own published Android figures of 7.9% and 9.3%. Repeating a setup changed nothing: two separately started runs of the German recording returned identical scores.

EFFORT

COMPONENT	SIZE	WHAT DROVE IT
Android built-in recogniser integration	L	This is a conversation with a system service rather than a library call. The interface can only be used from the main thread, every audio slice needs a fresh connection, and handing it a file instead of a microphone requires Android 13. The language pack has to be named, then installed through an API call, because the service does not expose its own download screen. Several defects appeared only on real hardware.
Whisper, run locally plus silence detection and keyword spotting	M	It arrives as a prebuilt binary with the inference runtime compiled in, so only the accelerators its publisher chose are available. One CPU accelerator works. The neural one is deprecated and unusable at this model's precision, and the GPU path needs the whole library rebuilt. Downloading and storing a 240 MB model belongs here too.
Measurement tooling scoring, statistics, test harness	XL	Two independent implementations of the scoring, held to each other by a test on the real recordings, plus the coverage check, CER, a significance test, and long runs that survive interruption. Audio import, run history and the comparison view belong to it. 487 automated tests keep it honest.

Sizes cover the whole job: the integration, the tooling each engine needed, and the defects that only appeared on hardware. They are not line counts and not hours. The measurement tooling was the largest single piece of work, and every finding below was invisible before it existed.

FINDINGS

- Recording length decides which engine to use. On the five short dialogues the two are within a point of each other, and the built-in recogniser wins the longest of them outright at 1.7% against 2.4%. The gap only opens on long-form audio, where Whisper leads by 2.0 points in English and 5.1 in German.
- Whisper is also the faster of the two, which is not obvious. It transcribes at roughly two and a half times real time, while the built-in service will not safely take audio faster than someone spoke it. A 21-minute recording takes Whisper 8 minutes and the built-in engine 21.
- The price of that is a 240 MB model shipped with the app and held in memory, and the price is real: two runs died part-way through when the system reclaimed the model, at 28% and 74% complete, with about 250 MB free. The built-in recogniser cannot fail this way, because the model belongs to the operating system.
- Naming the language pack matters more than any tuning. Left unspecified, the service quietly falls back to the phone's locale, here Indian English, a pack nobody chose and one it does not even list as installed. The second team measured 4.0% against 11.6% on identical audio from that alone.
- The Android speech service locks up. After roughly fifteen requests it pins a CPU core at 100% and stops answering, and the app then records the lost audio as empty text. Restarting the service at the first timeout cut the loss from four slices to one. Any long recording sent through it needs that watchdog.
- We can account for most of the built-in recogniser's remaining gap on German. Both runs lost the same passage of about 32 words, and putting it back brings them close to the second team's figure. The rest comes down to a different handset, a different speech-service version, and how the audio is divided: they send 17 minutes in one request, this app sends about 65 short ones.

WER, the word error rate, counts substituted, deleted and inserted words against the reference, using the alignment with the fewest errors. CER is the same measure over characters, and it separates formatting from genuine misrecognition: a high WER beside a low CER means the words were nearly right and the formatting was not, as in Whisper's 8.0% against 2.7% in English, where the difference is digits against spelled-out numbers. CER matters most in German, where compound words make the word-level rate read harsher than the transcript deserves.